

ADAM J. McFALL

5050 W Benedict Rd, Harrisburg, Missouri, USA 65256
(803) 979-3545 · amcfall96@gmail.com

EDUCATION



University of Georgia

Master of Science in Integrative Conservation & Sustainability

Athens, Georgia, USA

GPA: 3.71



University of South Carolina Aiken

Bachelor of Science in Biology

Aiken, South Carolina, USA

GPA: 3.858

GRANTS

Graduate Research Fellowship Program, National Science Foundation

Sep 2020 – Aug 2023

“Carryover Effects of Larval Stressors on Juvenile Gopher Frogs” - PI

\$34,000 annual living stipend, \$12,000 annual research stipend

Satch Krantz Conservation Fund, Riverbanks Zoo & Garden

Jan 2023

“A Soft Release Plan for Headstarted Gopher Frogs (*Rana capito*)” - PI

\$8,561

EXPERIENCE

Biologist

Sep 2023 – Present



Columbia Environmental Research Center | U.S. Geological Survey

Columbia, Missouri, USA

Series: 0401 - **Grade:** 7 (Term) | **Salary:** \$55,506.00 USD Per Year

40 hours/week

Assessment of fish communities in the Cuyahoga river before removal of a large dam

- Worked with stakeholders to conduct long-term multi-disciplinary environmental assessments of fish communities, water quality, and habitat variables along the Cuyahoga River and Gorge Metro Park reservoir prior to the Gorge Dam removal, resulting in the identification of thousands of fish across multiple species to pair with environmental variables
- Served as a crew member on electrofishing boats and barges, seine teams, and kayaked five 2–4 mile transects of the Cuyahoga River with radio and acoustic telemetry equipment, enduring low/high flow, heat, and low-class rapids without damaging equipment
- Conducted and managed water quality assessments using a velocimeter, handheld YSI, depth sounder, and rangefinder, resulting in water quality data that could be matched with fish community data to understand distribution patterns and population dynamics of species of greatest conservation need (SGCN)
- Created maps delineating the boundary of the Cuyahoga River in ArcGIS Pro for fish telemetry network integration and habitat evaluation
- Assisted with packing equipment, identifying project needs, and coordinated fieldwork for USGS team and partners in Ohio DNR, US EPA, Ohio EPA, and the National Park Service, resulting in clear procedures and roles for all parties involved

Testing an experimental approach for herding invasive carp in a backwater oxbow lake

- Coordinated and managed project fieldwork and served as a boat crew member, deploying 3500 ft of block net and 500 ft of corral net multiple times over a 2 mo window across two years.
- Drove boats and served as an electrofishing boat crew member to herd carp into large block net and corral net for invasive carp collection.
- Handled and processed hundreds of carp, set up sonar systems for monitoring catch, and communicated program management objectives with members of the public and state agency collaborators.
- Maintained a positive relationship with multiple members of the community including state agency partners, rowing club members, general public and recreationists.
- Retrieved thousands of feet of net with fish, untangled nets, mended holes in nets, and stowed nets in box crates on trailers.
- Comfortably backed in trucks with haul trailers and storage trailers at boat ramps with crowded conditions, and operated live-haul trailers for transporting fish.

Evaluating the use of cost-effective live-imaging sonar imagery to accurately measure fish

- Led project comparing size estimates of aquatic invasive species using research-grade sonar imagery, recreational-grade sonar imagery, and physical measurements in a pond setting, resulting in successful validation of sonar measurement techniques
- Automated file backup and upload processes from field computer to network folders using OneDrive and Windows task scheduler, allowing student service contractors and technicians to view data on their computers in near real-time
- Collated biological data in R, resulting in organized collection of physical measurements across all three techniques, which improved project workflow and prevented confusion among three technicians measuring fish
- Used statistical methods to determine appropriate sample sizes and determined project budget, schedule, and equipment needs to ensure timely completion amid personnel shortage
- Mentored two student service contractors and one technician by helping them prepare a presentation for a scientific meeting and teaching them how to run scientific trials with sonar equipment and use statistical methods to analyze data

- Ensured biological collection permit was up to date and served as an electrofishing crew member on two trips, resulting in the collection and transportation of 120 fish, with less than 2% mortality
- Demonstrated strong work ethic and gave feedback to principal investigator research staff on ways to improve project design and workflow

Developing live-imaging sonar surveillance arrays to monitor invasive carp passage

- Worked with federal and state agencies for an invasive carp prevention and early detection project to develop a live-imaging sonar surveillance array (LISSA) for monitoring invasive carp passage at Brandon Road Lock and Dam and Lock and Dam 5, supporting the president's objective of Protecting the Great Lakes from Invasive Carp
- Developed a technique using video captures devices (encoders) and YouTube live stream to record over 8 TB of continuous sonar imagery, allowing for real-time viewing of the imagery by lock operators as well as long-term imagery storage for later data analysis
- Conducted extensive computer and field testing, including overcoming obstacles related to computers, OBS Studio software, encoders, and Garmin sonar units to optimize recording file size and video quality
- Collated sonar data from each unit using automation processes in Windows and R and wrote multiple protocols and job aides for field assistants and inter-agency partners to check functionality and troubleshoot systems
- Created a budget and tracked equipment purchases in detailed inventory database to improve accountability and traceability
- Created protocols of procedures for collaborators and staff and provided technical direction for using equipment and software

Using acoustic telemetry and water quality to understand silver carp embayment use

- Collated biological (over 1.5 million fish detections) and environmental (over 10,000 water quality recordings) data for a long-term monitoring project using statistical methods in R, resulting in successful submission of a USGS data release to be published at a later date
- Conducted statistical analyses on large datasets and interpreted statistical outcomes to create study area and fish movement maps in ArcGIS and R, resulting in an oral presentation to be given at a scientific meeting
- Developed methods to extract local weather data from an API and pull in inter-agency partner fish tagging data, resulting in successfully matching these two datasets with existing water quality and fish detection data in R
- Deployed 42 acoustic telemetry receivers and 10 water quality sondes in backwater embayments along Kentucky Lake and Lake Barkley at Land Between the Lakes National Recreation Area in Kentucky to track silver carp movement over four winter sample periods (2021–2025)
- Calibrated and maintained remote multiparameter sondes and real-time sondes for water quality analysis
- Recorded global positioning system (GPS) data with a handheld Trimble unit for accurate deployment of receivers and sondes
- Backed up boat trailers, loaded boats and tied knots to secure acoustic receivers to concrete anchors prior to deployment

Conspecific feeding sounds as an attractant for invasive carp

- Served as a co-PI for acoustic telemetry study in experimental ponds, involving preparing the study plan proposal, determining schedule, project equipment, staffing, and budget needs
- Performed 10 surgeries and sutures to implant passive integrated transponder (PIT) and acoustic tags in grass carp and assisted with >100 more, with fewer than 5% mortality rate
- Operated complex HTI telemetry system, hydrophones, and Trimble GPS unit; conducted nightly sound exposure trials, resulting in consistent quality data collection
- Conducted data management and analysis on over 2.3 million fish detections using ArcGIS and R software, resulting in a draft publication and poster presentation for a scientific meeting
- Meticulously performed quality assurance/quality checks (QA/QC) of detection data to ensure quality product and kept a Quarto document of QA/QC procedures for other researchers to easily follow

Blue sucker tributary use using PIT tag array

- Restored five Biomark IS1001 master controllers to full functionality and built new PVC housings for 10 antennas resulting in timely delivery of equipment for graduate student thesis project studying blue sucker habitat use in Texas
- Presented progress updates and troubleshooting guide for inter-agency collaborators from state, university, and private agencies leading to accurate budget estimates and supply list for collaborators

Movement of Invasive Grass Carp in the Sandusky River

- Prepared acoustic receivers to be deployed in the Sandusky and Maumee Rivers to track grass carp movement as part of a long-term study
- Analyzed movement data with Fathom Position software to show how grass carp use areas in the Sandusky River and inform aquatic invasive species control programs
- Collated telemetry data using R and Microsoft Excel to upload to the Great Lakes Acoustic Telemetry Observation System (GLATOS) database to share telemetry data with inter-agency partners



Graduate Research Assistant
 Odum School of Ecology | University of Georgia
Salary: \$34,000.00 USD Per Year

Aug 2020 – Aug 2023
 Athens, Georgia, USA
 40+ hours/week

Implications of soft releasing for conservation of headstarted gopher frogs

- Designed a study to assess the effects of “soft releasing” (a delayed release strategy) on the movement, survival, and growth of the gopher frog (*Rana capito*)
- Constructed 18 3m x 3m pens to house gopher frogs and study behaviors prior to release—operated a mechanical trencher and brushsaw to dig lines for pen boundary and clear areas for pen construction

- Conducted forest plot delineations at two release sites to use for habitat evaluations and movement preference analyses, resulting in identifying high quality and low quality terrestrial plots
- Conducted radiotelemetry surveys on gopher frogs—fastened frogs with radiotransmitter belts
- Co-led a joint workshop for partners interested in gopher frog conservation from federal, state, university, and NGO agencies and gave tour of soft release field site, describing techniques and practices for management
- Created maps in QGIS and conducted data analyses in R and JMP software to evaluate movement differences between hard and soft release gopher frogs, resulting in a scientific paper currently in preparation
- Performed wetland delineations and habitat evaluations to identify potential breeding habitats and high quality terrestrial habitat, which resulted in confidently choosing a soft release site for gopher frogs
- Advised the US Forest Service on restoring breeding wetlands and potential dispersal wetlands to historic hydroperiods through tree-clearing and burning recommendations including seasonal timing and methods, resulting in a 10-year habitat management plan

Novel abnormalities in gopher frogs

- Characterized novel malformations in gopher frogs, resulting in a publication to a conservation journal
- Performed health screenings, photographs and radiographs, necropsies under microscope
- Developed a standardized protocol to keep wild gopher frogs in captivity for conservation purposes
- Became skilled at compiling, categorizing, and synthesizing scientific literature
- Assisted in writing animal use protocols approved by the UGA institutional animal care and use committee
- Presented work at multiple outlets for scientists and students

Utilizing carryover effects as a conservation tool in gopher frogs

- Designed mesocosm and field experiments to evaluate effects of snake predator cues on gopher frogs across life stages
- Assisted in husbandry of 250 larvae in mesocosms, involved long days in hot weather
- Modified a protocol for waterborne corticosterone extraction for gopher frogs
- Conducted water quality testing in mesocosms and in field setting with YSI – temperature, pH, dissolved oxygen



Life Science Technician

Savannah River Ecology Laboratory | University of Georgia

Salary: \$25,168.00 USD Per Year

May 2019 – Aug 2020
Aiken, South Carolina, USA
40 hours/week

Working with private landowners to collect environmental DNA (eDNA) and improve habitat management across South Carolina

- Traveled across the Savannah River Site and multiple private lands in South Carolina to collect eDNA and provide habitat management recommendations for SGCN
- Conducted wetland delineations in QGIS, habitat evaluations, and dipnet and visual surveys to identify breeding wetlands and recovery measures for endangered species
- Developed experience building partnerships with private land owners, touring their properties and offering counsel on land management topics related to species conservation, making sure to keep their values in mind and being respectful
- Became adept at sampling, extracting, and quantifying eDNA
- Attempted laboratory experiments aimed at estimating abundance of anurans in water using eDNA with Southern toad (*Anaxyrus terrestris*) and Southern leopard frog (*Lithobates sphenocephalus*) tadpoles

Density estimates of coyotes using genetic mark-recapture

- Drove thousands of miles of transects conducting visual encounter surveys for coyote (*Canis latrans*) scat on federal and private lands, 12 hrs driving each day
- Modified driving transects in Avenza Maps
- Extracted and quantified DNA from scat using Qubit and NanoDrop analyzers
- Became proficient with Avenza Maps, DNA extraction, PCR/qPCR, electrophoresis, JMP data analysis software

Waterborne corticosterone assays and hormone extractions from treefrogs

- Extracted hormones from Cuban treefrogs (*Osteopilus septentrionalis*) using a waterborne extraction method
- Modified a protocol for waterborne corticosterone extraction for Cuban treefrogs
- Became proficient at solid phase extractions and ELISAs, Gen5 spectrophotometry software



Department of Biology & Geology | University of South Carolina Aiken

Visiting Student Researcher

Salary: N/A

Sep 2018 – May 2019
Aiken, South Carolina, USA
15 hours/week

Undergraduate senior capstone project

- Conducted my senior thesis at the Savannah River Ecology Laboratory
- Designed and conducted feeding and hopping trials to analyze behavioral effects of the herbicide triclopyr
- Collected water samples from juvenile Cuban treefrogs (*Osteopilus septentrionalis*) to be used in hormone analysis
- Responsible for feeding, cleaning, and handling hundreds of Cuban treefrogs from larval to juvenile stages
- Assisted captive breeding effort by injecting adult Cuban treefrogs with reproductive hormones



University of South Carolina Aiken | Ruth Patrick Science Education Center Dec 2015 – May 2019
Animal Care Worker Aiken, South Carolina, USA
Salary: \$8.00 USD Per Hour 10 – 20 hours/week

Environmental outreach animal husbandry

- Responsible for reptile, amphibian, and bird husbandry, training, medicating, and cleaning
- Became skilled at handling screech (*Megascops asio*) and barred owls (*Strix vario*)
- Led outreach events for K-12 students and public involving handling and demonstrations of live animals

Planetarium

- Handled transactions of gifts and planetarium tickets, counted money and stocked items at planetarium store
- Organized activities for birthday events



Zoo Knoxville
Intern
Salary: N/A

Jun 2018 – Aug 2018
 Knoxville, Tennessee, USA
 16 hours/week

Herpetology

- Responsible for reptile and amphibian husbandry feeding, training, medicating, and cleaning
- Developed a research project to enhance welfare of captive Komodo Dragons (*Varanus komodoensis*)
- Assisted in breeding programs for multiple SGCN and critically endangered species including Louisiana pine snake (*Pituophis ruthveni*), bog turtle (*Glyptemys muhlenbergii*), spider tortoise (*Pyxis arachnoides*), Madagascar flat-tailed tortoise (*Pyxis planicauda*), and Roti Island snake-necked turtle (*Chelodina mccordi*)
- Gained experience collecting, incubating, and candling spider tortoise eggs
- Learned how to probe juvenile snakes to identify sex

Ornithology

- Responsible for bird husbandry feeding, training, medicating, and cleaning
- Worked with endangered species including African pygmy falcon (*Polihierax semitorquatus*)



PetSmart
Pet Care Associate
Salary: \$8.50 USD Per Hour

May 2018 – Aug 2018
 Knoxville, Tennessee, USA
 20 – 30 hours/week

- Responsible for feeding, handling, cleaning, rehabilitating reptiles, amphibians, birds, fish, and small mammals
- Store cashier and product stocking
- Became experienced with water quality tests and customer service
- Made animal care and husbandry recommendations to hundreds of pet owners

SCHOLARSHIPS & AWARDS

Star Award. US Department of the Interior & US Geological Survey. Columbia Environmental Research Center	2024
Best Oral Presentation Award, Rapid-Fire Session (1st place). SREL Graduate Student Symposium	2022
SSAR Student Travel Award (\$500). Society for the Study of Amphibians and Reptiles	2022
Best Undergraduate Platform Presentation (2nd place). SETAC North America	2019
Best Poster Presentation Award, Biology & Environmental Sciences Group F (2nd place). Discover USC	2019
SC LIFE Scholarship. University of South Carolina Aiken	2019
USCA Partnership Scholarship. University of South Carolina Aiken	2015 – 2019
USCA Bookstore Scholarship (\$315). University of South Carolina	2018
USCA Pep Band Scholarship. University of South Carolina Aiken	2015 – 2017
Aiken Rotary Club Scholarship (\$2000). Aiken Rotary Club	2015

PUBLICATIONS

- McFall A, Faulkner J, Mueller A, Byrd C, Calfee R. Conspecific Feeding Sounds as an Attractant for Invasive Carp. *In prep.*
- McFall AJ, Nelson KN, Grosse AM, Robinson PS, Henne JP, Lance SL. Improving translocation outcomes for an imperiled amphibian: comparing soft and hard release methods. *In prep.*
- McFall A, Ridgway J, Fischer J, Jones A, Deters J. Overwintering telemetry assessment of invasive Silver Carp in embayment habitats of Kentucky Lake and Lake Barkley, Kentucky, USA. USGS Data Release. *In review.*
- Stonecypher ET, Robinson PS, Nelson KN, McFall AJ. 2024. *Ambystoma opacum* (marbled salamander). Leucism. **Herpetological Review** 55(1):60–61.
- Vincent LM, Allender MC, Curtis AE, Garrison JC, Lance S, McFall A, Simmons A, Moorhead K, Adamovicz L. 2024. Health assessment of spotted (*Clemmys guttata*) and painted (*Chrysemys picta*) turtles in Cape Cod, Massachusetts, U.S.A, with detection of a novel adenovirus. **Journal of Zoo and Wildlife Medicine** 55 (3): 743–749.
- Nelson KN, McFall AJ, Stonecypher ET, Swartzbaugh CS, Allender MC, Lance SL. 2024. Development and validation of a scoring system for abnormalities in the Gopher Frog (*Rana capito*). **Herpetological Conservation and Biology** 19(1):117–128.

Vincent LM, Allender MC, Curtis AE, Madden NE, Cray C, Lance S, McFall A, Adamovicz L. 2023. Cutaneous myiasis and its relationship to wellness in Eastern Box Turtles (*Terrapene carolina carolina*) in Cape Cod, Massachusetts. **Journal of Zoo and Wildlife Medicine** 54(4):785–795

McFall AJ, Nelson KN, Stonecypher ET, Swartzbaugh CS, Allender MC, Burrell CE, Yabsley MJ, Lance SL. 2023. Morphological abnormalities in the Gopher Frog (*Lithobates capito*) during a head-starting event. **Herpetological Conservation and Biology** 18(3):436–449

McFall AJ, Ziemba J, Weir SM, Capps KA, Lance SL. 2023. Amphibian dispersal traits not impacted by triclopyr exposure during the juvenile stage. **Diversity** 15(2):215 special issue on “Amphibian ecology in geographically isolated wetlands”

REPORTS

McFall AJ, Lance SL, Grosse AM. A Soft Release Plan for Head-Started Gopher Frogs (*Rana capito*). Satch Krantz Conservation Fund Grant. 17 October 2023. Report.

PRESENTATIONS

McFall AJ, Ridgway JL, Fischer JR, Jones AC, Colvin M, Tompkins J. Evaluating silver carp aggregation behavior in large dynamic reservoirs to inform harvest. **American Fisheries Society Conference**. Columbus, Ohio, USA 3 Sep 2026. Oral.

McFall AJ. Improving project workflow and data management with Quarto. **Columbia Environmental Research Center Seminar Series**. Columbia, Missouri, USA 1 April 2026. Oral/Demo.

McFall AJ, Faulkner JD, Mueller AT, Byrd CG, Calfee RD. Evaluating Conspecific Feeding Sounds as an Attractant for Grass Carp. **Midwest Fish and Wildlife Conference**. St. Louis, Missouri, USA 21 January 2025. Poster.

McFall AJ, Nelson KN, Stonecypher ET, and Lance SL. Evaluating soft releasing as a component of headstarting programs for the gopher frog. **Columbia Environmental Research Center Seminar Series**. Columbia, Missouri, USA 7 February 2024. Oral.

McFall AJ. How abnormalities and soft releasing affect head-started gopher frogs and what it means for conservation. **Master’s Thesis Defense**. Aiken, South Carolina, USA 28 June 2023. Oral.

McFall AJ, Nelson KN, Stonecypher ET, and Lance SL. Soft releasing affects movement and behavior of head-started gopher frogs (*Rana capito*). **Southeast Partners in Amphibian and Reptile Conservation (SEPARC) Meeting**. Black Mountain, North Carolina, USA 24 February 2023. Oral.

McFall AJ, Nelson KN, Stonecypher ET, and Lance SL. Soft releasing affects movement and behavior of head-started gopher frogs (*Rana capito*). **Savannah River Ecology Laboratory Graduate Student Symposium**. Aiken, South Carolina, USA 21 October 2022. Oral.

McFall AJ, Nelson KN, Stonecypher ET, Swartzbaugh CS, Latham HN, and Lance SL. Abnormalities in the at-risk gopher frog (*Lithobates capito*): what we know and where we’re going. **Joint Meeting of Ichthyologists and Herpetologists**. Spokane, Washington, USA 29 July 2022. Oral.

McFall AJ, Nelson KN, Stonecypher ET, Swartzbaugh CS, and Lance SL. Developmental abnormalities complicate conservation of the Carolina gopher frog (*Rana capito*). **Odum School of Ecology Graduate Student Symposium**. Athens, Georgia, USA 12 February 2022. Oral.

McFall AJ. Assessing status and available habitat of the Mabee’s salamander (*Ambystoma mabeei*) in South Carolina. **GIS Applications for Natural Resources Final Presentations**. Athens, Georgia, USA 6 May 2021. Poster.

McFall AJ, Ziemba JL, Rodriguez G, and Lance SL. Effects of the herbicide triclopyr on the behavior and physiology of Cuban tree frogs. **SETAC North America 40th Annual Meeting**. Toronto, Ontario, Canada 4 November 2019. Oral.

McFall AJ, Ziemba JL, Rodriguez G, and Lance SL. Effects of the herbicide triclopyr on the behavior and physiology of Cuban tree frogs. **SREL Graduate Student Symposium**. Windsor, South Carolina, 19 July 2019. Poster.

McFall AJ, Ziemba JL, Rodriguez G, and Lance SL. Effects of the herbicide triclopyr on the behavior and physiology of Cuban tree frogs. **University of South Carolina’s Discover USC**. Columbia, South Carolina, 26 April 2019. Poster.

McFall AJ, Ziemba JL, Rodriguez G, and Lance SL. Effects of the herbicide triclopyr on the behavior and physiology of Cuban tree frogs. **Scholar Showcase**. Aiken, South Carolina, 12 April 2019. Oral.

McFall AJ and Lemyre C. Responsiveness of Komodo Dragons (*Varanus komodoensis*) to food, sensory, and social enrichment. **Zoo Knoxville Final Presentations**. Knoxville, Tennessee, 9 August 2018. Oral.

INVITED TALKS

McFall AJ, Nelson KN, Stonecypher ET, Swartzbaugh CS, and Lance SL. Obstacles and unexpected outcomes in Gopher Frog conservation research. **NC Congress of Herpetology**. Asheboro, North Carolina, USA 6 May 2023. Guest talk.

McFall AJ and Nelson KN. Gopher Frog conservation research: a story of unexpected challenges and paths forward. **Odum School of Ecology Freshwater Ecology Class**. Athens, Georgia, USA 5 October 2022. Guest lecture.

SPECIAL PROJECTS

A Soft Release Plan for Head-Started Gopher Frogs (*Rana capito*)

2023

- Lead a gopher frog reintroduction project in South Carolina, including conducting and managing project fieldwork with non-government agencies, state wildlife agencies, and university collaborators
- Wrote AUP and land use permits for project
- Purchase project materials, manage funding account, and bookkeeping
- Organize meetings with stakeholders to discuss goals, timeline, and key deliverables
- Coordinate site visits with property manager, South Carolina state herpetologist, USFWS, and Riverbanks Zoo personnel

- Scheduled pen construction involving stakeholders and volunteers and delegated responsibilities for members
- Operated mechanical trencher to outline pen boundary
- Installed 18 3m x 3m hardware cloth pens
- Radiotracked 42 hard release and soft released gopher frogs for a two month period in harsh conditions itemDrove truck and trailer down forest roads and service roads

Par Pond East (PPE) Land Management Plan

2022

Savannah River Ecology Laboratory

- Assisted in developing habitat management recommendations for the Savannah River Site's PPE region in South Carolina and assisted writing a 10-year management plan that the US Forest Service (USFS) will use
- Conducted wetland delineations and habitat evaluations to highlight habitat needs by individual stand and wetland in PPE for federal at-risk species and species listed by South Carolina Department of Natural Resources as high priority under the State Wildlife Action Plan (SWAP)
- Conducted site visits to groundtruth and evaluate current conditions of stands/wetlands and developed solutions to restore degraded stands/wetlands to suitable habitat
- Prioritized stands/wetlands into several categories to provide realistic management recommendations for USFS

Social, Economic, and Environmental Impacts of a Proposed Rock Quarry

2021

Hancock County, Georgia, USA

- Assisted Stack and Associates, P.C. in preparing evidence against a rock quarry establishment proposal by Mayfield, LLC in Hancock County, Georgia at the County Board of Commissioners Meeting
- Prepared a document highlighting the negative impacts of the proposed quarry on wildlife, community members, and historic sites
- County Board of Commissioners voted to deny the quarry permit to Mayfield, LLC by a 4-0 vote

RELEVANT COURSEWORK

Problems in Ecology

Aug 2022 – Dec 2022

University of Georgia

- Developed an original research project with the goal of soft releasing metamorphic gopher frogs at historic population in South Carolina
- Identified potential funding agencies for the project using web/database searches and personal communications
- Wrote a grant proposal outlining the soft release project
- Created budgets for research projects
- Gained experience with grant submission services at UGA including portal navigation and correspondence with grant specialists
- Submitted proposal to the Riverbanks Zoo & Garden's Satch Krantz Conservation Fund

GIS Applications for Natural Resources

Jan 2021 – May 2021

University of Georgia

- Developed a project to identify remaining suitable habitat in South Carolina for the Mabee's salamander (*Ambystoma mabeei*) and presented a poster as a final project, included management recommendations based on the proportion of suitable habitat versus available habitat
- Performed wetland delineations on dozens of wetlands across the range of the Mabee's Salamander in South Carolina to identify potential breeding sites and their threat from anthropogenic sources such as roads, noise, and degradation
- Became proficient at utilizing publicly available datasets for conservation purposes, including USGS land cover and hydrography, and USFWS National Wetlands Inventory, DOT public roads, and DNR Heritage Trust Program species records among others
- Conducted habitat evaluations for recreational/at-risk species, creating maps, raster/vector data, coordinate systems, statistics tables in ArcMap/ArcGIS

Environmental Law Practicum

Jan 2021 – May 2021

University of Georgia

- Partnered with Southern Environmental Law Center (SELC), tasked with identifying a vulnerable species in the Southeast US and developing a report outlining threats and conservation measures for it
- Contributed a detailed report that will be utilized for environmental justice cases by SELC in Virginia, North Carolina, and South Carolina for Mabee's salamander conservation, plus protection of isolated ephemeral wetlands not protected by the Clean Water Act and upland longleaf pine savanna management
- Assisted Stack and Associates, P.C. in preparing evidence against a rock quarry establishment proposal by Mayfield, LLC in Hancock County, Georgia, USA at the County Board of Commissioners Meeting
- Prepared a document that highlighted the potential impacts of the proposed quarry on wildlife, environmental justice concerns, and historic sites
- County Board of Commissioners voted to deny the quarry permit to Mayfield, LLC by a 4-0 vote
- Synthesized available information on wildlife impacts of per-and polyfluoroalkyl substances (PFAS) in 3 watersheds across GA/TN/AL for SELC
- Identified region-specific point-sources for PFAS along these watersheds and specific threats to species of concern

Herpetology
University of Georgia

Jan 2021 – May 2021

- Became proficient at identifying all reptiles and amphibians native to Georgia
- Conducted frog call surveys across Whitehall Forest in Athens, Georgia, USA for the USGS North American Amphibian Monitoring Program (NAAMP), used Kestrel Wind Meter 3000
- Responsible for monitoring a pond in Whitehall Forest for herp species from January-May using a variety of methods including minnow traps, turtle traps, PVC treefrog pipes, leaf litter bags, dipnet sweep surveys, eyeshine spotlighting, and call surveys
- Created a short video describing environmental DNA (eDNA) for conservation efforts of the gopher frog
- Developed class materials for an amphibian identification lab and led/taught other students

Ichthyology
University of South Carolina, Aiken

Jan 2018 – May 2018

- Learned principles in conducting scientific research in fishery
- Gained experience identifying fishes native to South Carolina
- Used backpack electrofisher to sample multiple stream communities to assess fish diversity

SKILLS

Certifications: GIS Applications in Fisheries; USGS Wilderness First Aid/CPR/AED; Electrofishing Safety

Skills: Experimental design; project management; data management; public outreach; wetland delineations; habitat evaluations; identifying wetland plants, organizing fieldwork; writing reports; relationship building; grant writing; data analysis; GIS; process automation; public speaking; statistical analysis; power tools; animal husbandry; boat operation; pulling and backing large trailers with trucks; training/mentoring young scientists and biologists; writing lab and field SOPs; fish surgery and suturing; animal husbandry, restraint, and training; surgical procedures on amphibians; hormone and DNA extractions; ELISAs; PCR/qPCR; electrophoresis; blood digestions; NMR/IR spectroscopy

Instrumentation/Equipment: Garmin GPSMAP 8610, GLS10 sonar module, and Livescope Plus LVS34, ARIS Explorer 1800, Vemco VR2AR and VR2Tx acoustic receivers, HTI model 290 ATR and hydrophones, Vemco EXO2 multiparameter and In-Situ Aqua TROLL sonds, handheld YSI, ATS R410 radiotelemetry receiver, LoTek radiotelemetry receiver, IS1001 Multiplexing Transceiver System and HPR Lite/Plus handheld PIT tag readers, Depstech endoscope, Kestrel 3000 Weather Meter, dissecting microscope, backpack electrofisher, Wildlife Acoustics recorders, Masterflex portable eDNA pump, Vermeer RTX200 pedestrian trencher, brushsaw and weedwacker

Software: *proficient* – Microsoft Office, R Studio, Quarto, JMP, QGIS/ArcGIS Pro, Google Earth, Avenza Maps, ArcGIS Field Maps, Garmin BaseCamp, Audacity, RavenPro, Fathom Position, BioTerm, AcousticTag, MarkTags, TagProg, Microsoft CoPilot *intermediate* – PyCharm, LabChart, ImageJ, Gen5

Computer Languages: *proficient* – R, Markdown, Pandoc, TeX/LaTeX, *intermediate* – Python, CSS, HTML

Former Member: The Wildlife Society Jul 2019 – Jul 2022, Nov 2023 – Nov 2024

Former Member: North Carolina Herpetological Society May 2023 – Dec 2023

Former Member: Society for the Study of Amphibians and Reptiles Jan 2022 – Dec 2023

Former Member: American Society of Ichthyologists and Herpetologists Jan 2022 – Dec 2022

Former Member: Society of Environmental Toxicology and Chemistry May 2019 – May 2022

LEADERSHIP & SERVICE

Educator 2026
Columbia Environmental Research Center 60th Anniversary Open House Columbia, Missouri, USA
Gave oral presentations and technology demonstrations to the public for sonar systems used in scientific research. Demonstrated the use of semi-autonomous kayaks for surveying waterbodies. Answered questions that the public had about research goals and methods.

Undergraduate Mentor 2025
Midwest Fish and Wildlife Conference St. Louis, Missouri, USA
Mentored a senior college student at the Midwest Fish and Wildlife Conference. The student was seeking out graduate school opportunities and interested in fisheries science.

SREL Diversity Committee 2022–2023
Savannah River Ecology Laboratory Aiken, South Carolina, USA
Responsible for planning initiatives to target improving racial and ethnic diversity within the SREL-community including special events and discussion groups.

GSS Executive Committee 2022
SREL Graduate Student Symposium Aiken, South Carolina, USA
Responsible for planning a symposium for graduate students, post-doctoral researchers, and technicians to present their research to the Savannah River Site community. Duties also included coordinating meetings, checking in on sub-committees, communicating event to SREL, and acting as liaison for an invited keynote speaker.

Founder's Fellow Mentor

2022

Joint Meeting of Ichthyologists & Herpetologists

Spokane, Washington, USA

Mentored a high school student at the Joint Meeting of Ichthyologists and Herpetologists. The student was the recipient of an award from the Society for the Study of Amphibians and Reptiles given to those who completed a Pre-College Scholars Program.

Reading Tutor

2022

UnitedWay Midland's Reading Consortium

Columbia, South Carolina, USA

Virtually tutored a kindergarten student struggling to read at grade-level for 45 minutes once per week. This program worked with majority-minority communities in the South Carolina midlands. I was able to engage my student in animal/habitat conservation through the books we read together and hopefully leave an imprint on them to consider a career in ecology.

Volunteer

2020–2021

EcoReach Backyard Critters

Athens, Georgia, USA

Volunteered for the platform/logistics team for EcoReach's Backyard Critters program. This was an initiative to reach K-12 students to spark a fascination for wildlife. Students submitted photos of animals or plants they found with questions to the Backyard Critters Submission Page. I was responsible for helping develop this page for students and teachers to post their findings and answer questions.

Volunteer Research Assistant

2020; one day event

Georgia Adopt-A-Stream Program

Jackson, South Carolina, USA

Collected baseline water quality data and invertebrate diversity counts along transects of Hollow Creek (tributary of the Savannah River) in the Silver Bluff Audubon Center and Sanctuary in partnership with the Augusta-Aiken Audubon Society.

Volunteer Educator

2019; one day event

Savannah River Ecology Lab's Touch an Animal Day

Aiken, South Carolina, USA

Volunteered for Savannah River Ecology Lab's (SREL) Touch an Animal Day (TAAD), providing information to visitors on amphibian behavior, ecology, and threats to habitat for each species.

Judge's Assistant

2018, 2019; one day event

Future City Competition

Aiken, South Carolina, USA

Collaborated with USCA and Savannah River Site (SRS) faculty who judged middle school student projects. I was responsible for timing and giving the introductory remarks for each school to the judges before students presented 3-dimensional city models centered around energy efficiency and environmental preservation.

Student Liaison

2016–2018

Pacer Pulse Basketball Band at USCA

Aiken, South Carolina, USA

Responsible for proposing and securing budget funds to the USCA Student Government Association (SGA), cheer design, vocal initiation, encouraging positive performance, and attending required leadership meetings.

Volunteer Musician

2015–2018

Aiken Youth Orchestra and Aiken Civic Orchestra

Aiken, South Carolina, USA

Volunteered to play as a fill-in principal horn musician for seasonal concerts in both Aiken youth and Aiken civic orchestras.